

No. of Printed Pages : 4

220125/170025/120025

Roll No.

/030025

2nd Sem / Agri, Automobile, Mechanical, Mechanical Tool & die Design) Chem, P&P, Civil, Elect, Plastic, Prod, Mechatronics, GE CAD/CAM, CNC, Metallurgy, F&F, Civil Constr, Pack Tech, Printing Tech, Power Stat Engg, Power Eltx, Elect & Eltx Engg, Paint Tech, Ruber Tech, Polymer Engg., Highway Engg., Fab Tech. Fire Tech & Safety, AME

**Subject : Mechanical Engineering Drawing -I /
Engineering Drawing - II**

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 Which thread is used in lead screw of lathe?

- | | |
|------------------|-------------------|
| a) B.S.W. thread | b) Square thread |
| c) Acme thread | d) Knuckle thread |

Q.2 Width across flats in hexagonal nut is

- | | |
|------------------------|---------|
| a) $1.5D + 3\text{mm}$ | b) $3D$ |
| c) $1.8D + 9\text{mm}$ | d) $2D$ |

Q.3 Halving and bridle joints are

- | | |
|------------------|-------------------|
| a) welded joints | b) riveted joints |
| c) cotter joints | d) wooden joints |

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- Q.4 Coupling may be broadly divided into
- a) rigid coupling b) flexible coupling
 - c) Both (a) and (b) d) flange coupling
- Q.5 The innermost portion of thread is called
- a) crest b) root
 - c) flank d) slope
- Q.6 The standard taper of the taper key is
- a) 1:10 b) 1:100
 - c) 1:1000 d) All of the above

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 Define assembly drawing.
- Q.8 Define flank.
- Q.9 Define foundation bolt.
- Q.10 Define lead.
- Q.11 Define machine screw.
- Q.12 Define stud.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Show the B.A thread by rough sketch.
- Q.14 Sketch the Split pin nut.
- Q.15 Draw freehand sketch of Rag bolt.
- Q.16 Define sunk key and list its types.
- Q.17 Draw one view of square nut when internal diameter of nut is 20 mm.
- Q.18 Draw free hand sketch of Muff coupling.
- Q.19 Explain caulking and fullering process.
- Q.20 Draw the following studs.
- a) Stud with square neck
 - b) Stud with collar
- Q.21 What are the differences between single and multiple start threads?
- Q.22 Define cotter and write the names of cotter joints.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Draw sectional Elevation and Top view of double riveted lap joint (chain type). Take diameter of rivet = 20 mm.

Q.24 Draw the free hand sketch of a non-protected flange coupling with proportional dimensions.

Q.25 Detail drawing of closed mortice and tenon joint is shown in figure 1. Draw the following assemble views:

- a) Front view
- b) Side view
- c) Top view

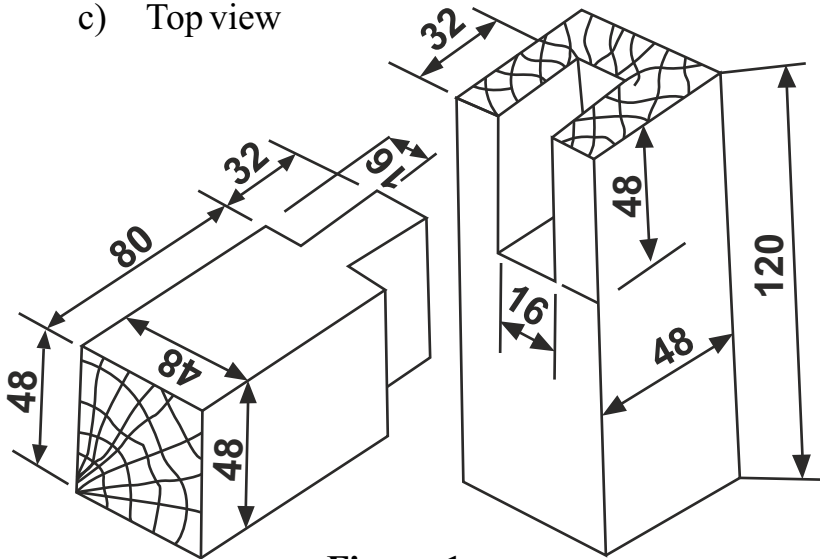


Figure-1

- Note • Dimension units may be taken in mm.
- Assume suitable dimension if not specified or clearly visible in the given figure.